Peiwen Li

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Education

Tsinghua University Beijing, China
Master, Data Science and Information Technology, GPA: 3.93/4 (Major 4/4) *Sep. 2022 - Jun. 2025*
Advisors: Prof. Wenwu Zhu and Prof. Yang Li
Main Courses: Machine Learning(4.0); Fundamentals of Digital Image and Video Processing(4.0)

Minzu University of China Beijing, China
Bachelor's degree, Statistics, GPA: 4.43/5 (94.43/100), Rank 1st *Sep. 2018 - Jun. 2022*
Main Courses: Mathematical Analysis(99); Advanced Algebra(92); Probability Theory(97); Applied Mathematical Statistics(99); Data Structure(94); Principle Operational Research; Applied Stochastic Processing

Research Interests

Machine Learning, especially in Large Language Models, Causal Inference and Graph Neural Networks.

Publications

1. **Peiwen Li**, Xin Wang, Zeyang Zhang, Yijian Qin, Ziwei Zhang, Jialong Wang, Yang Li, Wenwu Zhu.
   Causal-Aware Graph Neural Architecture Search under Distribution Shifts.
   Under review at NeurIPS 2024. link
2. **Peiwen Li**, Xin Wang, Zeyang Zhang, Yuan Meng, Fang Shen, Yue Li, Jialong Wang, Yang Li, Wenwu Zhu.
   RealTCD: Temporal Causal Discovery from Interventional Data with Large Language Model.
   Under review at CIKM 2024. link
3. **Peiwen Li**, Yuan Meng, Xin Wang, Fang Shen, Yue Li, Jialong Wang, and Wenwu Zhu.
   Causal Discovery in Temporal Domain from Interventional Data.
   The 32nd ACM International Conference on Information and Knowledge Management (CIKM), 2023. link

Experience

Multimedia and Network Big Data Lab, Tsinghua University Beijing, China
Advisors: Prof. Wenwu Zhu and Prof. Xin Wang. *Nov. 2021 - Now*

Causal Molecular Relational Learning via GNN-LLM Co-curriculum | *Independent Research, Ongoing*

Causal-Aware Graph Neural Architecture Search under Distribution Shifts | *Independent Research*
- Propose superior approach CARNAS, which is able to capture the causal graph-architecture relationship during the architecture search process and discover the generalized graph architecture under distribution shifts.
- The first to study graph neural architecture search under distribution shifts from the causal perspective.

RealTCD: Temporal Causal Discovery from Interventional Data with Large Language Model|*Independent Research*
- Propose the efficient RealTCD, a framework able to leverage domain knowledge through LLMs to discover temporal causal relationships without interventional targets, which is especially applicable in industrial scenarios.
- The first to solve the problem with Large Language Models (LLMs) and without interventional targets.

Causal Discovery in Temporal Domain from Interventional Data | *Independent Research*
- Propose novel TECDI, the first attempt to employ interventional data for real-world temporal causal discovery.
- Construct the first-ever real-world temporal interventional dataset from a data center of a cloud service company.

Explainability in Graph Neural Architecture Search by Decorrelated Subgraph Reweighting|*Undergraduate Thesis*
- Design a framework e-GraphNAS to analyze why a certain architecture is obtained by a GraphNAS algorithm.

Alibaba Cloud, Alibaba Group Hangzhou, China
Research Intern. Alibaba Innovative Research Program. *May 2023 - Now*
- Anomaly Detection based on GNN and Causal Discovery, and applications in AIOps of data center.

Misc

Skills: English: CET6: 594, CET4: 607. Coding: C, Python, R, Matlab, SPSS, SAS, LaTeX.
Academic Service: Conference Reviewer: KDD(2024)

Awards

- **Comprehensive Excellence Scholarship**, Tsinghua SIGS *2023*
- **Outstanding Graduate Award of Beijing** (4%) *2022*
- **Merit Student of Beijing** (1%) *2021*
- **Meritorious Winner in Interdisciplinary Contest In Modeling** *2020*